Dr. Torsten John

Postdoctoral Researcher — Max Planck Institute for Polymer Research

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RESEARCH INTERESTS

Self-Assembly ▪ Peptide Fibrils ▪ Functional Amyloids ▪ Membranes ▪ DNA Nanotechnology ▪ Nanointerfaces ▪ Protein Corona ▪ Surface Functionalization ▪ Nanomedicine ▪ Modelling and Simulation

EDUCATION

PhD (Dr. rer. nat.), Chemistry, Leipzig University, *summa cum laude (1.0)* — 2015–2020

MSc, Chemistry, Leipzig University, *excellent (1.1)* — 2012–2015

BSc, Chemistry, Leipzig University, *very good (1.5)* — 2009–2012

Grading scheme is 1.0-4.0 with 1.0 being the highest grade.

PROFESSIONAL EXPERIENCE

Postdoctoral Researcher — Jun 2023–present
Max Planck Institute for Polymer Research, Mainz, Germany

Postdoctoral Researcher — Nov 2020–May 2023
Department of Biological Engineering,
Massachusetts Institute of Technology (MIT), Cambridge, MA, USA

Postdoctoral Researcher — Feb 2020–Oct 2020
Leibniz Institute of Surface Engineering (IOM), Leipzig, Germany

Graduate Research Assistant *during PhD Program* — Jul 2015–Feb 2020
Leibniz Institute of Surface Engineering (IOM), Leipzig, Germany

Visiting Research Fellow *during PhD Program* — Jan 2018–Jul 2018
School of Chemistry, Monash University, Melbourne, Australia

Visiting Research Fellow *during PhD Program* — Aug 2016–Mar 2017
School of Chemistry, Monash University, Melbourne, Australia

Summer Research Fellow *during Master Program* — Dec 2014–Feb 2015
School of Chemistry and Molecular Biosciences,
The University of Queensland (UQ), Brisbane, Australia

Visiting Research Fellow *during Master Program* — Feb 2013–Jul 2013
School of Chemistry, Monash University, Melbourne, Australia

Visiting Research Fellow *during Master Program* — Aug 2012–Sep 2012
School of Applied Sciences, RMIT University, Melbourne, Australia

Undergraduate Research Intern — Aug 2011–Sep 2011
Helmholtz Centre for Environmental Research (UFZ), Leipzig, Germany

Industry Intern — Jul 2009–Aug 2009
SKW Stickstoffwerke Piesteritz GmbH, Wittenberg, Germany

FELLOWSHIPS AND AWARDS

Research Fellowships

Feodor Lynen Return Fellowship, Alexander von Humboldt Foundation, Germany — 2023–2024
Feodor Lynen Research Fellowship, Alexander von Humboldt Foundation, Germany — 2021–2022
PhD Fellowship, Friedrich Ebert Foundation (FES), Germany — 2016–2019
Endeavour Research Fellowship, Australian Government, Australia — 2018
Summer Research Fellowship, The University of Queensland (UQ), Australia — 2014–2015
DFG Research Fellowship, Graduate School BuildMoNa, Leipzig University, Germany — 2012–2013
DAAD ISAP Fellowship, German Academic Exchange Service, Germany — 2013
DAAD RISE Worldwide Fellowship, German Academic Exchange Service, Germany — 2012

Awards

Finalist, Leibniz Dissertation Award, Leibniz Association, Germany — 2021

Junior Researcher Award, Leibniz Institute of Surface Engineering (IOM), Germany	2020
Finalist, European Young Chemist Award (EYCA), EuChemS, Europe	2018
Element (Pb) Award, Periodic Table of Younger Chemists, IUPAC, International	2018
Early Career Researcher Award, Royal Australian Chemical Institute (RACI), Australia	2016
GDCh Graduate Award, German Chemical Society (GDCh), Germany	2009

Competitive Programs

IUPAC/GDCh Young Observers Program, IUPAC World Chemistry Congress, Netherlands	2023
Young Scientist, 70th and 71st Lindau Nobel Laureate Meeting, International	2020–2022
Graduate School, DFG CRC TRR 102 iRTG "Polymers: random coils and beyond", Germany	2015–2020
Young European Talent (YET) Program, Maastricht, Netherlands	2019
SciFinder/CAS Future Leaders Program, Columbus and Washington, D.C., USA	2017
Life Science College Member, German Academic Scholarship Foundation, Germany	2013–2015
NESACS-GDCh Student Exchange, Boston and New Haven, USA	2013
Participation, German Selection for International Chemistry Olympiad (IChO), Germany	2008–2009

Scholarships and Travel Grants

GDCh Travel Grant, 47th IUPAC World Chemistry Congress, Paris, France	2019
GDCh Travel Grant, 7th EuChemS Chemistry Congress, Liverpool, UK	2018
e-fellows Scholarship, ZEIT and McKinsey & Company, Germany	2012–2018
Study Scholarship, Friedrich Ebert Foundation (FES), Germany	2012–2015

PUBLICATIONS

Total: 22 (Scientific: 19, Viewpoints: 3), **Citations: 506**, **h-index: 12** (Google Scholar, 20 March 2024)
First author: 15, Corresponding author: 5, Journal covers: 4

Scientific Publications

(19) S.O. Aderinto, **T. John**, A. Onawole, R.P. Galleh, J.A. Thomas*, Iridium(III)-based minor groove binding complexes as DNA photocleavage agents, *Dalton Trans.* (**2024**) 10.1039/D4DT00171K.

(18) M. Hayn#, **T. John**#, J. Bandak, L. Rauch-Wirth, B. Abel*, J. Münch*, Hybrid Materials From Peptide Nanofibrils and Magnetic Beads to Concentrate and Isolate Virus Particles, *Adv. Funct. Mater.* (**2024**), 2316260.

(17) **T. John***, A. Rampioni, D. Poger, A.E. Mark*, Molecular Insights into the Dynamics of Amyloid Fibril Growth: Elongation and Lateral Assembly of GNNQQNY Protofibrils, *ACS Chem. Neurosci.* 15 (**2024**), 716–723. (Cover Page)

(16) **T. John***, L.L. Martin, B. Abel*, Peptide Self-Assembly into Amyloid Fibrils at Hard and Soft Interfaces—From Corona Formation to Membrane Activity, *Macromol. Biosci.* 23 (**2023**), 2200576.

(15) **T. John***, S. Piantavigna, T.J.A. Dealey, B. Abel, H.J. Risselada, L.L Martin*, Lipid oxidation controls peptide self-assembly near membranes through a surface attraction mechanism, *Chem. Sci.* 14 (**2023**), 3730–3741. (Cover Page)

(14) **T. John**, L.L. Martin, H.J. Risselada*, B. Abel*, Curvature model for nanoparticle size effects on peptide fibril stability and molecular dynamics simulation data, *Data Brief* 45 (**2022**) 108598.

(13) **T. John**, J. Adler, C. Elsner, J. Petzold, M. Krueger, L.L. Martin, D. Huster, H.J. Risselada*, B. Abel*, Mechanistic insights into the size-dependent effects of nanoparticles on inhibiting and accelerating amyloid fibril formation, *J. Colloid Interface Sci.* 622 (**2022**) 804–818.

(12) X. Wang#, S. Li#, H. Jun#, **T. John**, K. Zhang, H. Fowler, J.P.K. Doye, W. Chiu*, M. Bathe*, Planar 2D Wireframe DNA Origami, *Sci. Adv.* 8 (**2022**) eabn0039.

(11) H. Jun, X. Wang, M.F. Parsons, W.P. Bricker, **T. John**, S. Li, S. Jackson, W. Chiu, M. Bathe*, Rapid prototyping of arbitrary 2D and 3D wireframe DNA origami, *Nucleic Acids Res*. 49 (**2021**) 10265–10274.

(10) **T. John**, J. Bandak, N. Sarveson, C. Hackl, H.J. Risselada, A. Prager, C. Elsner, B. Abel*, Growth, Polymorphism, and Spatially Controlled Surface Immobilization of Biotinylated Variants of $IAPP_{21-27}$ Fibrils, *Biomacromolecules* 21 (**2020**) 783–792.

(9) **T. John**, G.W. Greene, N.A. Patil, T.J.A. Dealey, M.A. Hossain, B. Abel, L.L. Martin*, Adsorption of Amyloidogenic Peptides to Functionalized Surfaces Is Biased by Charge and Hydrophilicity, *Langmuir* 35 (**2019**) 14522–14531. (Cover Page)

(8) **T. John**, T.J.A. Dealey, N.P. Gray, N.A. Patil, M.A. Hossain, B. Abel, J.A. Carver, Y. Hong, L.L. Martin*, The Kinetics of Amyloid Fibrillar Aggregation of Uperin 3.5 is Directed by the Peptide's Secondary Structure, *Biochemistry* 58 (**2019**) 3656–3668.

(7) **T. John**[#], A. Gladytz[#], C. Kubeil, L.L. Martin, H.J. Risselada, B. Abel*, Impact of nanoparticles on amyloid peptide and protein aggregation: a review with a focus on gold nanoparticles, *Nanoscale* 10 (**2018**) 20894–20913. (Cover Page)

(6) **T. John***, B. Abel, L.L. Martin, The Quartz Crystal Microbalance with Dissipation Monitoring (QCM-D) Technique Applied to the Study of Membrane-Active Peptides, *Aust. J. Chem.* 71 (**2018**) 543–546.

(5) L.L. Martin*, C. Kubeil, S. Piantavigna, T. Tikkoo, N.P. Gray, **T. John**, A.N. Calabrese, Y. Liu, Y. Hong, M.A. Hossain, N. Patil, B. Abel, R. Hoffmann, J.H. Bowie, J.A. Carver, Amyloid aggregation and membrane activity of the antimicrobial peptide uperin 3.5, *Pept. Sci.* 110 (**2018**) e24052.

(4) **T. John**, T. Thomas, B. Abel, B.R. Wood, D.K. Chalmers, L.L. Martin*, How kanamycin A interacts with bacterial and mammalian mimetic membranes, *Biochim. Biophys. Acta - Biomembr.* 1859 (**2017**) 2242–2252.

(3) **T. John**, Z.X. Voo, C. Kubeil, B. Abel, B. Graham, L. Spiccia, L.L. Martin*, Effects of guanidino modified aminoglycosides on mammalian membranes studied using a quartz crystal microbalance, *Med. Chem. Commun.* 8 (**2017**) 1112–1120.

(2) A. Gladytz, **T. John**, T. Gladytz, R. Hassert, M. Pagel, S. Naumov, H.J. Risselada, A.G. Beck-Sickinger, B. Abel*, Peptides@mica: From affinity to adhesion mechanism, *Phys. Chem. Chem. Phys.* 18 (**2016**) 23516–23527.

(1) M. Pagel, R. Hassert, **T. John**, K. Braun, M. Wießler, B. Abel, A.G. Beck-Sickinger*, Multifunctional Coating Improves Cell Adhesion on Titanium by using Cooperatively Acting Peptides, *Angew. Chem. Int. Ed.* 55 (**2016**) 4826–4830.

Viewpoints and Policy Publications

(3) **T. John***, K.E. Cordova, C.T. Jackson, A.C. Hernández-Mondragón, B.L. Davids, L. Raheja, J.V. Milić*, J. Borges*, Engaging Early-Career Scientists in Global Policy-Making, *Angew. Chem. Int. Ed.* 62 (**2023**) e202217841.

(2) **T. John**[#], M. Cieślak[#], D. Vargová, S.M. Richardson, V. Mougel*, J. V. Milić*, The Role of Early-Career Chemists in European Policy-Making, *Chem. - A Eur. J.* 27 (**2021**) 6359–6366.

(1) C.A. Urbina-Blanco*, S.Z. Jilani*, I.R. Speight*, M.J. Bojdys*, T. Friščić*, J.F. Stoddart*, T.L. Nelson, J. Mack, R.A.S. Robinson, E.A. Waddell, J.L. Lutkenhaus, M. Godfrey, M.I. Abboud, S.O. Aderinto, D. Aderohunmu, L. Bibič, J. Borges, V.M. Dong, L. Ferrins, F.M. Fung, **T. John**, F.P.L. Lim, S.L. Masters, D. Mambwe, P. Thordarson, M. Titirici, G.D. Tormet-González, M.M. Unterlass, A. Wadle, V.W.-W. Yam, Y. Yang, A diverse view of science to catalyse change, *Nat. Chem.* 12 (**2020**) 773–776.

* Corresponding author, [#] Equal contribution

TEACHING AND MENTORING

Co-Facilitator, Leadership and Professional Strategies and Skills (8.396 / 8.397) 2022
Massachusetts Institute of Technology (MIT), Cambridge, MA, USA
Co-facilitated lectures, led table discussions, graded homework

Kaufman Teaching Certificate, Massachusetts Institute of Technology (MIT) 2021
Completed program to develop teaching and learning skills, including microteaching

Instructor, Advanced Physical Chemistry (13-111-0441-N / 13-221-043) 2016
Leipzig University, Leipzig, Germany
Managed and supervised lab experiment, conducted oral pre-exams, graded protocols

Teaching Associate, Medicinal Chemistry (CHM3930 / BTH1011) 2013
Monash University, Melbourne, Australia
Assisted students in completing problem sets during workshops, graded homework

Workshop Facilitator, Chemistry Seminars for Olympiad Participants 2011–2012
Friends of the Chemistry Olympiad (FChO), Leipzig, Germany
Prepared workshop material, facilitated workshops, assisted students

Teaching Assistant, Inorganic Chemistry (13-111-0211-N) 2010–2011
Leipzig University, Leipzig, Germany
Assisted students in completing experiments during laboratory classes

Mentoring

Mentored as supervisor or co-supervisor undergraduate students and interns at Monash University (3), the Leibniz Institute of Surface Engineering (1), and at MIT (1)

CONFERENCE & COLLOQUIUM PRESENTATIONS

Conference Talks

(13) SupraLife Second School, Aveiro, Portugal, 2024
(12) 49th IUPAC World Chemistry Congress, The Hague, Netherlands, 2023
(11) MRS Fall Meeting, Boston, MA, USA, 2022
(10) ACS Fall National Meeting, Chicago, IL, USA, 2022
(9) ACS Fall National Meeting, Chicago, IL, USA, 2022 (2nd talk)
(8) 48th IUPAC World Chemistry Congress, virtual, 2021
(7) ACS Virtual Postdoc Symposium, virtual, 2020 (**invited**)
(6) ACS Fall Meeting, virtual, 2020
(5) 7th Minisymposium of the SFB-TRR 102, Leipzig, Germany, 2019 (**invited**)
(4) 7th EuChemS Chemistry Congress (ECC), Liverpool, UK, 2018 (**invited**)
(3) 7th EuChemS Chemistry Congress (ECC), Liverpool, UK, 2018 (2nd talk)
(2) RACI Peptide Users Group Symposium, Melbourne, Australia, 2016 (**invited**)
(1) ACS Northeast Regional Meeting (NERM), New Haven, CT, USA, 2013

Conference Posters

(15) RSC Poster Twitter Conference, virtual, 2023
(14) RSC Poster Twitter Conference, virtual, 2022
(13) EMBO Workshop "Designing functional biomolecular assemblies: Beyond biology", virtual, 2021
(12) RSC Poster Twitter Conference, virtual, 2020
(11) 47th IUPAC World Chemistry Congress, Paris, France, 2019
(10) 21st JCF Spring Symposium, Bremen, Germany, 2019
(9) International Discussion Meeting on Polymer Crystallization (IDMPC), Wittenberg, Germany, 2017
(8) GDCh Science Forum, Berlin, Germany, 2017
(7) SciFinder Future Leaders Program, Columbus, OH, USA, 2017 (**invited**)
(6) 19th Postgraduates' Conference on Chemistry, Heraklion, Greece, 2017
(5) 19th JCF Spring Symposium, Mainz, Germany, 2017
(4) 5th Portuguese Young Chemists Meeting, Guimarães, Portugal, 2016
(3) 18th JCF Spring Symposium, Kiel, Germany, 2016
(2) 12th European Biological Inorganic Chemistry Conference (EuroBIC), Zürich, Switzerland, 2014
(1) GDCh Science Forum, Darmstadt, Germany, 2013

Colloquium and Seminar Talks

(5) School of Chemical Sciences, University of Auckland, Auckland, New Zealand, 2023 (**invited**)
(4) Research School of Chemistry, Australian National University, Canberra, Australia, 2023 (**invited**)

(3) Department of Chemistry, Johannes Gutenberg University, Mainz, Germany, 2022 (**invited**)
(2) Laboratory Prof. Dr. Wah Chiu, Bioengineering, Stanford University, virtual, 2021 (**invited**)
(1) Institute of Medical Physics and Biophysics, Leipzig University, Leipzig, Germany, 2019 (**invited**)

PROFESSIONAL MEMBERSHIPS

German Chemical Society (GDCh), since 2009
American Chemical Society (ACS), Community Associate, since 2020
Alumni Association, Faculty of Chemistry and Mineralogy, Leipzig University, since 2011

SERVICE AND LEADERSHIP

Reviewer of Scientific Manuscripts (*direct invitation*)
Data in Brief, iScience, Journal of Visualized Experiments, Nature Communications, Soft Matter, Trends in Chemistry

Conference Organization

Advisory Board & Scientific Committee Member, IUPAC World Chemistry Congress, Netherlands	2023
Session Co-Chair, MRS Fall Meeting, Boston, MA, USA	2022
Advisory Board Member, CRF-ChemCYS, Belgium	2022
Session Co-Chair, IUPAC World Chemistry Congress, France	2019
Session Chair, Minisymposium of the SFB-TRR 102, Germany	2019
Co-Organizer, European Young Chemists' Meeting / Delegate Assembly, Germany	2019
Co-Organizer, Young Chemists Crossing Borders Exchange (YCCB), UK/USA	2018

Leadership in Early-Career Networks and Scientific Societies

Past Chair and Advisor, International Younger Chemists Network (IYCN)	Aug 2023–present
Advisory Board, International Sustainable Chemistry Collaborative Centre (ISC3)	Nov 2022–present
Selection Committee, EuChemS, Historical Landmarks Award	May 2018–present
Task Group Member, Global Conversation on Sustainability (IYCN, IUPAC)	Feb 2022–Dec 2023
Chair, International Younger Chemists Network (IYCN)	Aug 2022–Aug 2023
Vice-Chair, International Younger Chemists Network (IYCN)	Aug 2021–Aug 2022
Leadership Team, ChemVoices.org (IYCN, IUPAC)	May 2020–Dec 2021
Committee Leader, International Younger Chemists Network (IYCN)	Mar 2019–Aug 2021
Co-Leader, Round Table Leipzig, Assoc. for Chemistry and Economics (VCW)	Mar 2019–Apr 2020
Committee Member, International Younger Chemists Network (IYCN)	Apr 2017–Mar 2019
Secretary, European Young Chemists' Network (EYCN)	May 2017–Mar 2019
Editorial Board Member, Chemistry in Europe Newsletter, EuChemS	May 2017–Mar 2019
Webmaster, Regional Chapter, German Young Chemists' Network (JCF)	Jun 2010–Mar 2019
Treasurer, German Young Chemists' Network (JCF)	Sep 2014–Sep 2017
Vice-Chair, Regional Chapter, German Young Chemists' Network (JCF)	Jun 2010–Oct 2014

Service in Students' Representation

Scholars Representative, Leipzig, Friedrich Ebert Foundation (FES)	Nov 2013–Nov 2014
Advisor, Chemistry Student Council, Leipzig University	Jun 2012–Jan 2013

SCIENCE COMMUNICATION AND OUTREACH

Outreach Publications *(not counted in Publication Summary)*

(5) M.E.A. Dilanas, E. Dobbelaar, C. Gerischer, A. Haseloer, **T. John**, C. Neumeier, Interview with Prof. Dr. Benjamin List – Nobel Laureate in Chemistry 2021, *Chem. - Eur. J.* (**2022**) e202201236.
(4) **T. John**, The Ins and Outs of a Ph.D. Across Continents, *ChemViews* (**2018**).
(3) **T. John**, Scientists Must Challenge the Status Quo, *ACS Axial* (**2018**).
(2) **T. John**, Neue Ansätze gegen Alzheimer, *Chemie Unserer Zeit*. 49 (**2015**) 361.
(1) V. Koester, M. Linden, A. Augustin, **T. John**, C. Schrapel, Working for the JCF – The Young Chemists' Forum of the German Chemical Society, *ChemViews* (**2015**).

Talks and Panel Discussions

(6) IYCN Webinar, "The Role of Young Scientists in Shaping Global Policies", virtual, 2023 (**invited**)
(5) ISC3 Global Week of Sustainable Chemistry, "People, Plastic and Pollution - How Can Youth Engagement Solve the Global Challenges of Today?", virtual, 2021 (**invited**)
(4) ChemVoices Webinar, "Fostering Innovation through Collaboration", virtual, 2020 (**invited**)
(3) 47th IUPAC World Chemistry Congress, "Young Chemists' Networks and their Impact on Education, Research and Society", Paris, France, 2019
(2) Three Minute Thesis (3MT), "From Sunscreen to Alzheimer's Disease: A Story about Peptide Aggregation", Leipzig University, Leipzig, Germany, 2019
(1) EuChemS Workshop at the European Parliament, "Chemical Sciences for Horizon Europe, education and employability", Brussels, Belgium, 2019 (**invited**)

Public Outreach

Support of Public Outreach activities of the IYCN in my role as Vice-Chair and Chair	2021–2023
Jury Panel Member, EYCN Video competition "Chemistry Rediscovered"	2017–2019
Jury Panel Member, EYCN Photo competition "Photochimica"	2017
Coordinated chemistry experiment booths, Science Nights, Leipzig, Germany	2010–2014
Invited speakers for popular science talks, Leipzig, Germany	2010–2014

MEDIA COVERAGE

(3) EYCN Podcast, How to Organize your Postdoc, *Spotify* (**2021**).
(2) M. Ferns, SciMeetings Spotlight: Torsten John, *ACS Axial* (**2021**).
(1) L. Wang, 2017 SciFinder Future Leaders, *C&EN* 95 (**2017**) 40-42.

PROFESSIONAL SKILLS & COMPETENCIES

Languages

German (native), English (fluent)

Completed Professional Development *(selected)*

Science Communication, Klaus Tschira Foundation	2022
Future Faculty Workshop, Northeastern University	2021
HPC and Data in Materials Design and Discovery, Intersect Australia	2021
Leadership and Professional Strategies and Skills (LEAPS), MIT	2021
European Science Diplomacy, S4D4C	2020
Responsible Conduct of Research, CITI Program	2020
Leading Without Authority, American Chemical Society (ACS)	2019
Reviewer Lab, American Chemical Society (ACS)	2018
Computational Structural Biology with ROSETTA, Vanderbilt and Leipzig University	2016